

ARUSHI SANJAY AGRAWAL

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SUMMARY

Machine Learning Engineer with 3+ years shipping production ML and AI systems end-to-end, and an MSc in Machine Learning from UCL (Distinction). From research and model development through distributed pipelines to deployment at scale, with prior experience at large technology organisations. A computer vision and 3D perception specialist, currently one of two ML engineers at a pre-seed startup, owning its entire defect-detection and point-cloud pipeline from model architecture to production serving.

EDUCATION

UNIVERSITY COLLEGE LONDON

Master of Science, Machine Learning - Distinction

London, UK

2024 - 2025

- Thesis: Enhancing Spatial Reasoning in VLA models, Huawei
- Courses: Probabilistic and Unsupervised Learning, Bayesian Deep Learning, Artificial Intelligence for Sustainable Development, Applied ML, Applied DL, Statistical Natural Language Processing, Open-Endedness and General Intelligence, Computer Vision

INDIAN INSTITUTE OF TECHNOLOGY GUWAHATI (IITG)

Bachelor of Technology, Engineering Physics, GPA - 8.45/10.0

Guwahati, India

2018 - 2022

- Thesis: Handling Bias in Visual Question Answering
- Courses: Linear Algebra, Multivariate Calculus, Statistics, Real and Complex Analysis, Computing, Mathematical Physics, Computational Physics, Probability & Statistical Mechanics, Pattern Recognition & Machine Learning

PROFESSIONAL EXPERIENCE

INFRAMIND LABS

Machine Learning Engineer

London, UK

Oct 2025 - Present

- Own end-to-end computer-vision and point-cloud ML pipeline as one of two ML engineers at a pre-seed startup, turning raw 3D point clouds of rail tunnels into actionable structural insights for 7+ clients on an interactive 3D inspection platform.
- Lifted 3D defect recall 27.6% and image-level recall to 91.3% by researching, training and fine-tuning multiple RGB-D instance-segmentation models (YOLO, Mask2Former) with a custom recall-weighted loss and a region-of-interest input channel.
- Created a generalised seven-class dataset and an internal benchmark to score current and future models: a multiview 3D protocol replacing a 2D metric stuck flat at 0.61 across every variant.
- Accelerated production workflows by up to 88% and raised cluster parallelism from 0.97x to 13.5x, profiling each stage end to end and migrating Prefect orchestration to pure Ray.
- Run team MLOps, keeping CPU and GPU workloads healthy on Azure Kubernetes, shipping a FastAPI serving layer through CI/CD with autoscaling, and cutting storage costs 60%.

HUAWEI

Reinforcement Learning Intern

London, UK

Apr 2025 - Sep 2025

- Exposed memorisation failure in vision-language-action foundation models (OpenVLA, $\pi 0$) through a robustness protocol of multi-seed perturbation experiments in simulation, collapsing 80% and 98% benchmark success to 0%.
- Restored spatial reasoning in frozen, imitation-learned VLA policies for robotic manipulation, boosting task success from 0% to 17.3% with a training-free scene-graph adaptor (open-vocabulary detection, SAM, CLIP/BLIP, XMem tracking).
- Selected each of six perception stages by benchmarking 25+ models across controlled ablation experiments, against relaxed-success and multi-seed distance metrics designed to isolate perceptual grounding from motor-skill limits.
- Fine-tuned a Qwen2.5-VL vision-language model with LoRA/PEFT on synthetic data, routing to avoid catastrophic forgetting.
- Deployed a multimodal pipeline as a real-time FastAPI microservice at 70 ms per refresh in a closed-loop control stack.

ADOBE

Software Development Engineer (GenAI Team)

Bangalore, India

Jul 2022 - Aug 2024

- Engineered and integrated multi-modal machine learning features into Adobe Premiere Pro (30M+ users), leveraging C++ and MVVM architecture to streamline video editing workflows, increasing user satisfaction.
- Optimised SegNeXt-based model for video mask generation using ONNX, improving accuracy by 17%.
- Analysed Hadoop-sourced user interaction data, driving a 13% increase in feature adoption through data-driven insights.

TEXT MERCATO SOLUTIONS

Machine Learning Intern

Remote

Sep 2021 - Mar 2022

- Implemented U2Net architecture for image segmentation and Boundary-Aware PoolNet for salient object detection.
- Enabled automated product description generation using Named Entity Recognition and LSTMs, reducing inference latency by 50%.

PROJECTS

RETRIEVAL-AUGMENTED LANGUAGE AGENTS FOR GAME PLAY

Jan 2025 - Apr 2025

Open Endedness and AGI / Prof. Tim Rocktaschel, UCL

[GitHub](#), [Paper](#)

- Built a LLM-based AI agent on RAG and a FAISS vector database, adding a context-summarization step pushing benchmark progression from 0.4 to 0.7 (+75%) over an unsummarized baseline.
- Trained a reward-ranking model with RLHF and a decision transformer for long-term action evaluation.

MEMORY-AUGMENTED CRITIQUE FOR SELF-IMPROVEMENT IN LLMS

Jan 2025 - Apr 2025

Statistical Natural Language Processing / Prof. Pontus Saito Stenetorp

[GitHub](#), [Paper](#)

- Developed a multi-agent system with Super Tiny Language Models (77M parameters) with LoRA, prompt engineering, structured debate, and memory modules, reaching a ROUGE-1 F1 score of 0.406 in dialogue summarisation.

TECHNICAL SKILLS

Programming Languages: Python, C++, JavaScript/ TypeScript, HTML, CSS, MATLAB, R, SQL, LaTeX

Machine Learning Frameworks: PyTorch, TensorFlow, Keras, Scikit-learn, HuggingFace

Libraries & Tools: NumPy, Pandas, OpenCV, Matplotlib, FAISS, CUDA, Git, MySQL, LangChain, Copilot, FastAPI, Docker, Kubernetes / AKS, Azure, CI/CD

LEADERSHIP & ACHIEVEMENTS

- **Advanced C++ Boot Camp, Adobe:** Ranked 1st by designing and implementing a high-performance application.
- **Shadowfax Cascade Cup 2022:** Secured 1st place out of 1,086 participants in a national Data Science Quiz.
- **Joint Entrance Exam (JEE) 2018:** Achieved 99.5 percentile among 1.26 million candidates in India.
- **Academic Mentor 2021-2022:** Guided team of 6 students in academics and career planning.